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New Late Eocene pole for Australia, time-averaging of remanence directions, and palaeogeographic reference systems

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SUMMARY

A new Late Eocene (~36.5 Ma) South palaeomagnetic pole based, on 33 sampling levels in the Browns Creek Formation of the Otway Basin, south-eastern Australia, lies at 65.5°S, 112.5°E with $A_{95} = 2.5°$. This pole fills a 30 Myr gap in the Palaeogene part of the Australian apparent polar wander pole path, and appears well suited for calibrating the path because of a well-defined age and evidence that the sequence was magnetized during a depositional period long enough to thoroughly average out secular variations. The remanence seems to have been acquired during post-depositional re-working of the sediment. The pole agrees with one biostratigraphically dated and three undated poles that constrain the Palaeogene trajectory.

All Palaeogene poles on the path lie west of an alternative apparent polar wander path. The latter is modelled by low-order curve fitting of much more widely scattered poles from basalts in which the scatter appears to be due to incomplete time-averaging of remanence directions. The new pole also reinforces a previously observed systematic westward displacement, by a few degrees, of the early Tertiary palaeomagnetic poles relative to hotspot-derived poles. This displacement may be caused by true polar wander and/or departures from the geocentric axial dipole model. It again highlights the approximateness of the available palaeogeographic reference systems.

Key words: APW (apparent polar wander), Australia, palaeogeography.

INTRODUCTION

The Cainozoic apparent polar wander path (APWP) for Australia reflects a progressive northward drift of the continent from moderately high southern latitudes in the Cretaceous period. It provides a method for age-dating and a reference for studying the movement of terranes on the margins of the Australian plate. Most parts of the APWP are spatially well defined by poles from sediments, basalts and poorly age-controlled laterites (Idnurm 1985). The time-scale for the APWP is, however, fixed by only five dated poles, placing a high reliance on the precision and accuracy of each pole. Least constrained is the Palaeogene part of the APWP which contains only two poles, one at 26 Ma near the beginning of the period and the other at 58 Ma near the end (Idnurm 1985)—a time gap of about 30 Myr.

The present study aimed to fill the Palaeogene gap with a Late Eocene pole from the Browns Creek Formation of the Otway Basin. This sedimentary formation satisfies two conditions essential for precise time-calibration: a well-defined age and a remanence acquired over a sufficiently long period to record a time-averaged field. The second requirement is not satisfied by rocks such as single lava flows or groups of flows extruded during short episodes of volcanism, which record the direction of the geomagnetic field over brief time intervals (e.g. McElhinny 1973). These volcanics give an estimate of the position of the geomagnetic pole, which is subject to secular variations and is, therefore, unsuitable as a stable geographic reference.

Palaeomagnetism assumes that, over a sufficiently long period, the time-averaged geomagnetic pole (i.e. the palaeomagnetic pole) coincides approximately with the geographic pole. (The agreement is generally unlikely to be better than a few degrees because the time-averaged geomagnetic field is not perfectly dipolar; e.g. Merrill & McElhinny 1983.) For Australia, the amplitude of secular variations is comparable to the net Tertiary apparent polar wander. For example, the present geomagnetic field (declination 5°, inclination −56°) at Alice Springs near the centre of the continent, would yield a geomagnetic pole that is displaced from the geographic pole by 14°—nearly half the angular span of the Cainozoic APWP. Therefore, for the accuracy usually required in Cainozoic applications,

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the averaging has to be thorough, making it is essential to use rocks that contain long and continuous records of the geomagnetic field. The relatively small scatter of the Australian laterite-derived poles about a common trajectory (Idnurm 1985) implies efficient time-averaging and indicates a prolonged magnetization in laterites (Idnurm & Schmidt 1986). Similarly, the slow deposition of the clays of the Browns Creek Formation (<1 cm Kyr^{-1}, see below) is likely to give effective time-averaging at almost the 2 cm scale of the individual specimen.

GEOLOGY

The Late Eocene Browns Creek Formation was deposited in a shallow-marine environment (Shafik & Idnurm, in preparation) at two small, fault-generated embayments of the Otway Basin. Parts of the succession crop out at two localities on the Victorian coast west of Cape Otway, at Browns Creek and Castle Cove (Fig. 1). In combination, these outcrops span the complete formation of about 40 m thickness. The basal 10 m are *Turritella aldingae*-rich clays (Fig. 2), red-brown near the contact with the underlying Palaeocene Johanna River Sand and grey elsewhere. The *Turritella* clays are overlain by about 2 m of glauconitic sand, then 20 m of bryozoal marls, and lastly by 10 m of light grey clays. Deposition is believed to have taken place during a continent-wide transgression (McGowran 1989).

A Late Eocene age is indicated for the basal *Turritella* clays by the nannofossils *Cyclicargolithus reticulatus, Isthmolithus recurvus, Chiasmolithus oamaruensis*, and the foraminifer *Acarinina collactea* (Shafik & Idnurm, in preparation). A combination of magneto- and biostratigraphy (Shafik & Idnurm, in preparation) suggests that the *Turritella* clays are within Chrons 16–18.

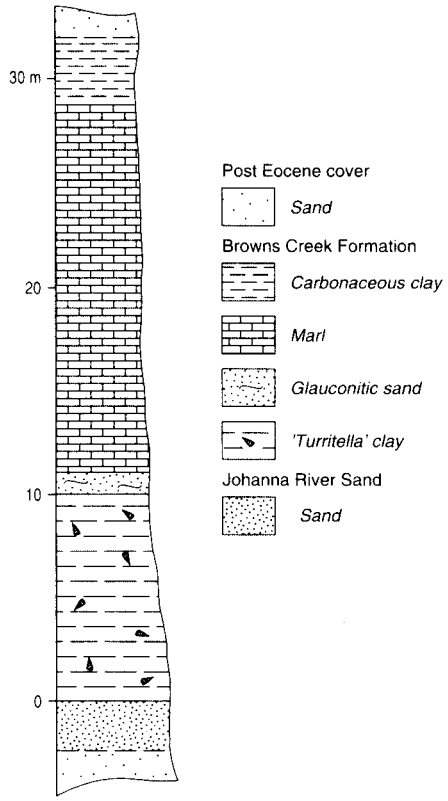


Figure 2. Stratigraphic section exposed at the sampling site. The base and top of the section are covered by Pliocene and Recent sands. Stratigraphic levels are relative to the base of the Browns Creek Formation. For the reasons discussed in the text, sampling was restricted to the *Turritella* clays.

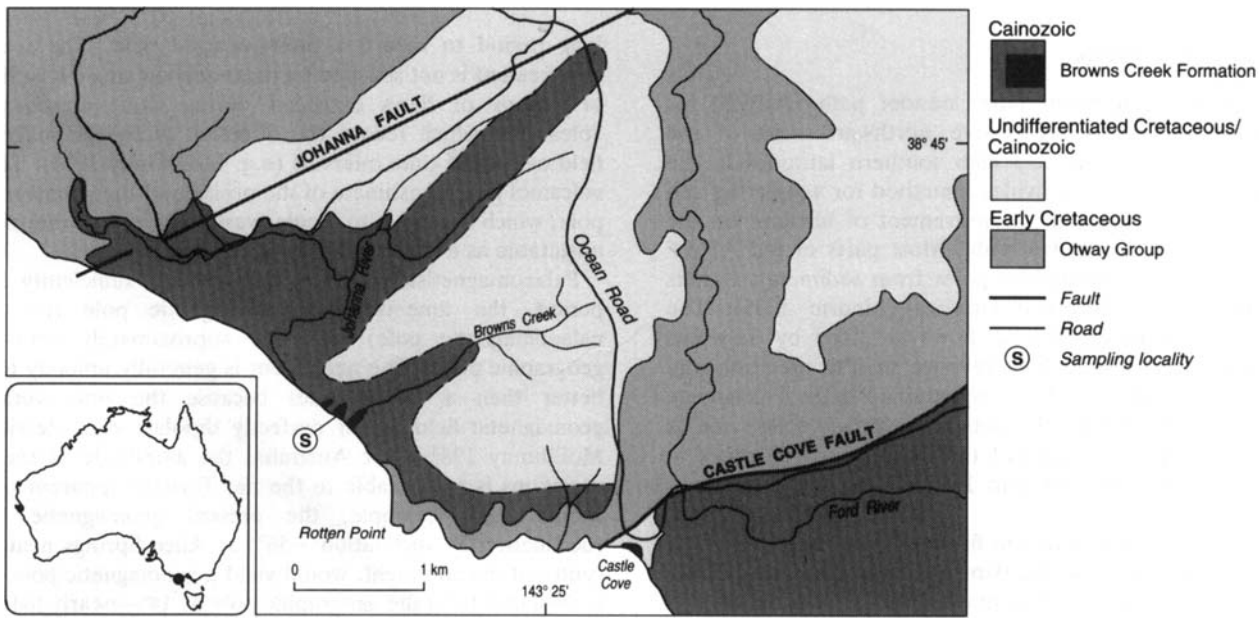


Figure 1. Geological sketch map (modified from Singleton 1973) of the Browns Creek Formation outcrop region. S locates the sampling site.

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SAMPLING AND MEASUREMENTS

Preliminary sampling of the Browns Creek Formation yielded stable remanence from only the basal *Turritella* clays. That sequence, which outcrops as a coastal cliff near Browns Creek (Fig. 1), was therefore selected for the present study. Since the clays are poorly consolidated, the samples were collected in plastic cube-containers by a carving technique (Nott, Idnurm & Young 1991) or, occasionally, by forcing the container into the sediment, the first technique being preferred because the application of force could distort the sediment (Gravenor, Symons & Coyle 1984). The cubes were oriented with a magnetic compass and an inclinometer, the local magnetic deviation being obtained by comparison of magnetic and sun compass readings at the outcrop.

In all, 105 samples were collected, usually two from each level (details in Shafik & Idnurm, in preparation). The average separation between the levels was 18 cm, the stratigraphic heights being obtained with a Jacobs rod and an Abney level. The site has a uniform bedding tilt of 8° towards 326°N, determined by Abney level sightings on prominent strata within the bryozoal marls.

The measurements were made at the Black Mountain Palaeomagnetic Laboratory, Canberra, using cryogenic magnetometers. Thermal demagnetization to 200 °C was used in conjunction with rock magnetic measurements. Otherwise the specimens were demagnetized in alternating fields, in a two-axis Schonstedt tumbler-demagnetizer, because of the low melting temperature of the plastic sample containers. The field increments were 5 mT or less up to 60 mT, and thereafter 10 mT up to a maximum field of 100 mT. Remanence directions were obtained by principal component analysis (Kirschvink 1980).

RESULTS

The initial intensities of remanence in the *Turritella* clays ranged from 0.5 to 3 mA m^{-1} in the basal 4 m and from 1 to 30 mA m^{-1} above. The greater variability and general increase of intensity above 4 m were accompanied by a reduction of scatter in the orthogonal projection plots (Fig. 3b). However, no change in the remanence directions was observed across the 4 m level to suggest time differences in magnetization and, therefore, the possibility of overprinting.

With few exceptions, the remanences contained two components (Fig. 3). The lower coercivity component with negative inclination and northerly declination demagnetized at peak fields of 10–25 mT and is interpreted as a recent overprint. The second, higher-coercivity component is interpreted as primary. In magnetically stable specimens this component was nearly extinct at 100 mT, the highest field of the demagnetizer.

A remanence direction was included in the pole calculation if its mean angular deviation (MAD) in the principal component analysis was less than 5°. Sixty-six directions from 33 sampling levels qualified. The directional means of these levels form two tight and anti-parallel groups (Fig. 4) for which the McFadden & McElhinny (1990) reversal test is positive at a 90 per cent confidence level (classification B). The overall mean direction gives the Late Eocene pole 112.5°E, 65.5°S (confidence angle $A_{95} = 2.5°$). Table 1 summarizes the results.

MAGNETIC PROPERTIES

Acquisition of isothermal remanent magnetization (IRM) in the clays was studied to help find the cause of change in the intensity and stability of the remanence at the 4 m level.

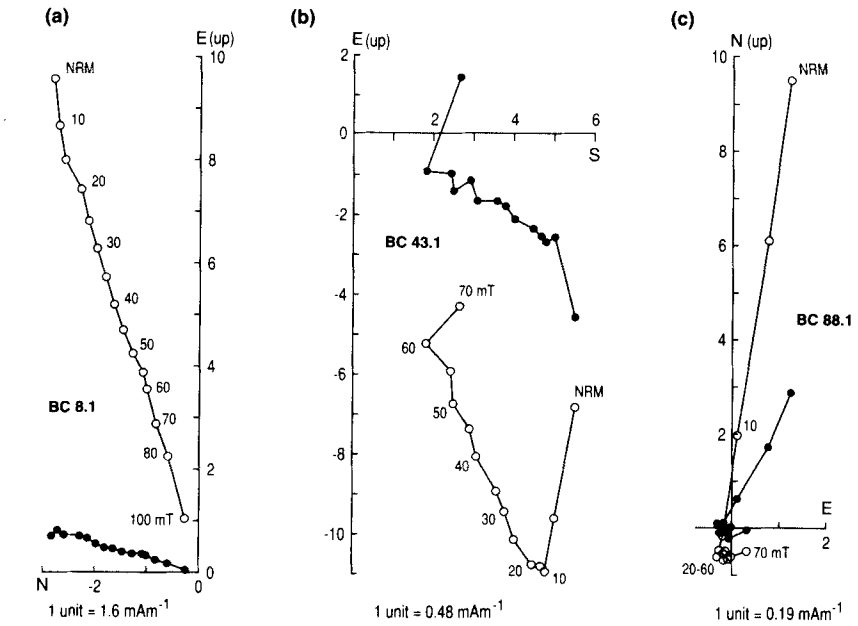


Figure 3. Examples of orthogonal projection plots for AF demagnetization illustrating: (a) typical behaviour of samples above the 4 m level; (b) and (c) onset of instability, common below the 4 m level, at higher demagnetizing fields. Solid (open) symbols indicate projection onto horizontal (vertical) plane. Directions are bedding corrected.

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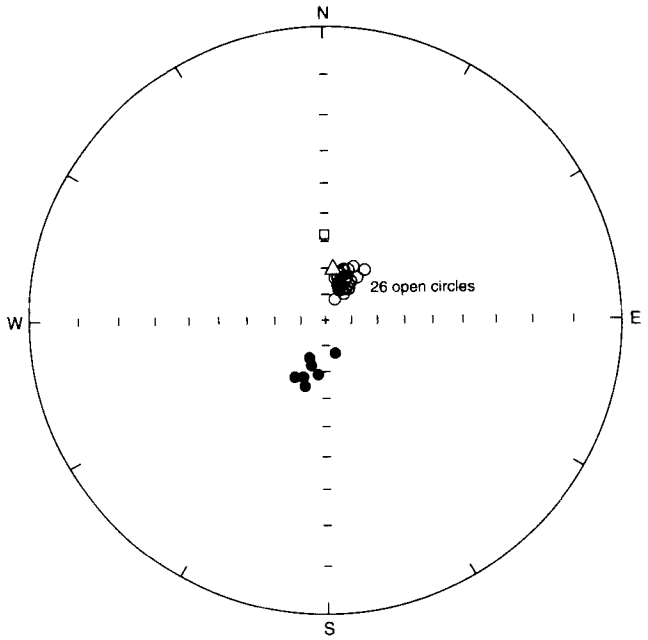


Figure 4. Bedding-corrected mean remanence directions for 33 sampling levels in *Turritella* clays. Open (solid) symbols indicate directions in upper (lower) hemisphere. The square and the triangle show the present dipole and present geomagnetic field directions respectively. Note that the angular difference between the present dipole and geomagnetic field is comparable to the net apparent polar wander since the Eocene, emphasizing the need to eliminate secular variation effects in Cainozoic palaeomagnetic poles. Equal angle projection.

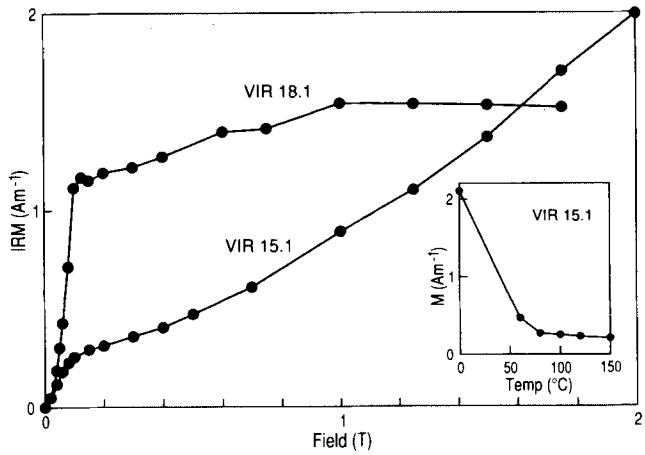


Figure 5. IRM acquisition curves for sample VIR 18.1 from above and sample VIR 15.1 from below the 4 m level in the Browns Creek Formation. Insert shows thermal demagnetization characteristics of IRM for sample VIR 15.1.

Fig. 5 shows the acquisition characteristics for two samples, one from below and the other from above the 4 m level. Both samples contain a low- and a high-coercivity component.

The low-coercivity components start to saturate at about 0.1 T and are, therefore, probably due to magnetite (Dunlop 1972). Indeed, magnetite is abundant in the surrounding Early Cretaceous volcaniclastics of the Otway Group. In sample VIR 18.1 from above 4 m, the low-coercivity component dominates the IRM, whereas in sample VIR 15.1 from below 4 m it is subordinate.

Thermal demagnetization eliminated the high-coercivity component in sample VIR 15.1 at about 80 °C (Fig. 5, insert). The combination of high coercivity and low unblocking temperature suggests goethite as the remanence carrier in that sample. Therefore, the marked increase in strength and stability of the magnetic signal above 4 m appears to be due to a change from goethite- to magnetite-dominated remanence. The directional consistency (Fig. 4) suggests that the remanences of the two minerals were acquired in the same general period, so the difference in mineralogy probably reflects a change of either provenance or weathering environment, rather than some process that may reset the magnetization well after deposition.

Table 1. Summary of palaeomagnetic results.

	Remanence Direction						South Pole			
	In Situ		Bedding Corrected							
N(n)	Dec. (deg.)	Inc. (deg.)	Dec. (deg.)	Inc. (deg.)	k	α_{95} (deg.)	Lat. °S	Long. °E	K	A_{95} (deg.)
Normal Polarity										
26(57)	8.2	-68.0	26.0	-73.1	384.5	1.5	64.0	112.5	144.4	2.4
Reverse Polarity										
7(9)	183.8	64.0	196.9	69.8	147.4	5.0	71.1	112.3	57.1	8.1
Combined Polarities										
33(66)	7.1	-67.2	23.8	-72.4	262.7	1.6	65.5	112.5	99.9	2.5

Angle between normal and reverse polarity mean directions: 4.6°. Critical angle for positive fold test, classification B (McFadden & McElhinny 1990): 5.2°. Site co-ordinates: 143.39°E, 38.78°S. *N* is the number of sampling levels (unit weight); *n* is the number of specimens. *K* and *k* are the precision parameters (Fisher 1953), and α_{95} and A_{95} are the semi-angles of 95 per cent confidence.

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DISCUSSION

Apparent polar wander

Figure 6 shows the pole, BC, from the Browns Creek Formation in relation to the Australian APWP (Idnurm 1985). BC lies about 3° west of the mean trajectory. Its position between the 26 Ma and 58 Ma poles is consistent with the Late Eocene age of the formation.

The high precision of BC may be attributed to a long period of magnetization of each specimen. As discussed by Shafik & Idnurm (in preparation), the ~10 m thick *Turritella* clays are estimated from combined bio- and magnetostratigraphy to have been deposited over a period longer than 1 Myr. Therefore, the average 2 cm specimen would have acquired its remanence in more than 2000 years—long enough to smooth out much of the secular variation in the geomagnetic field. We note that the remanence in the clays is likely to be post- rather than syn-depositional because of sediment disturbance by fauna and winnowing bottom currents: the burrowing *Turritella* is abundant in the clays and bedding structures are absent. These post-depositional processes are likely to delay the magnetization without affecting its rate of acquisition.

The age of the pole is estimated from the mean stratigraphic level (6.1 m) of the samples used in the pole calculation. That level corresponds to an age of 36.5 Ma on the time-scale of Cande & Kent (1992) (Shafik & Idnurm, in preparation).

While the Browns Creek Formation pole is consistent with the APWP shown in Fig. 6, it disagrees with an APWP

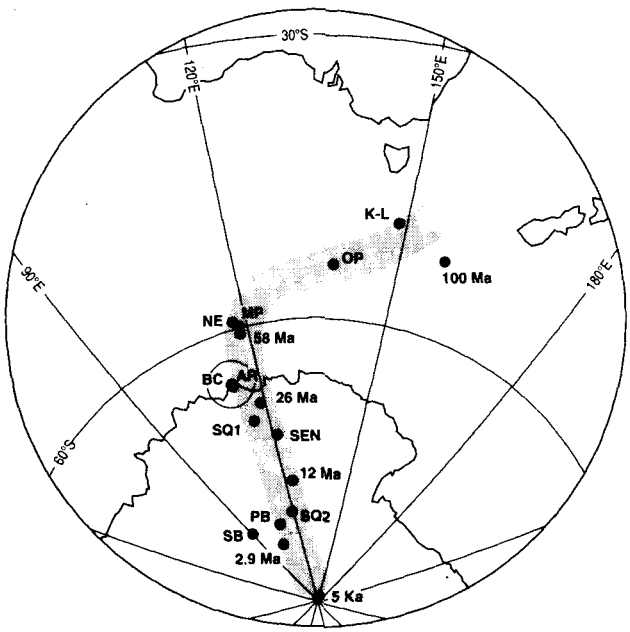


Figure 6. Pole BC from the Browns Creek Formation plotted on the Late Mesozoic–Cainozoic APWP for Australia (after Idnurm 1985). OP is a Cretaceous thermal overprint pole in south-eastern Australia. The other acronyms denote poles from undated laterite profiles: AR: Senior *et al.* (in preparation); MP: Idnurm & Senior (1978); NE: Schmidt & Ollier (1988); SB: Schmidt, Currey & Ollier (1977); SEN, SEQ1, SEQ2: Idnurm (1985); PB: Schmidt & Embleton (1976).

obtained by low-order curve fitting of much more scattered poles from basalts (Musgrave 1989). The latter APWP, referred to here as the basalt-derived APWP, is based on six poles selected from a larger volcanic data set. The two APWPs are compared in Fig. 7(a). Pole BC (labelled 36.5 Ma) lies about 5° west of the basalt-derived APWP. Similarly, the 58 Ma pole and the laterite poles MP and NE lie west of the APWP, as does the pole AR from ferruginous weathering in the Hale Basin of Central Australia (Senior *et al.*, in preparation). It seems unlikely that all five poles have similar westward errors. As discussed previously (Idnurm 1985), at least some of the scatter of the basalt poles is likely to be caused by an incomplete averaging-out of palaeosecular variations. The statistical treatment, therefore, seems unable to eliminate the effects of residual palaeosecular variations in the pole path. For dating purposes, a 5° difference on the Australian APWP is equivalent on average to an age difference of 10 Myr in the Cainozoic.

As shown in Fig. 7(b), the pole BC (36.5 Ma) reinforces evidence for a westward displacement of the Early Tertiary poles relative to poles calculated from the hotspot model of Duncan (1981; see Musgrave 1989). While the displacement is too small to be supported by the individual palaeomagnetic and hotspot-derived poles, it emerges from the general trends. The difference in the paths also receives support from new hotspot results of Jenkins & Packham (1993). Differences between the hotspot and palaeomagnetic frameworks are well known and have been attributed partly to true polar wander (e.g. Besse & Courtillot 1991); indeed, true polar wander appears to have been rapid during the Late Cretaceous—Early Tertiary (Besse & Courtillot 1991). Contributions to the difference may come also from non-dipole components in the long-term average geomagnetic field. A large non-dipole component in the Eocene geomagnetic field has recently been suggested from European data (Westphal 1993).

The palaeomagnetic–hotspot discrepancies again highlight the approximate nature, due to departures from idealized models, of the two palaeogeographic reference systems when used as absolute palaeogeographic references. These approximations would affect especially the accuracy of palaeolatitude estimates. In some applications, such as age dating, an absolute geographic system is not essential. However, even there substantial errors are likely to arise when one reference system is used as the proxy for the other on the assumption that the two systems link precisely through the absolute geographic reference frame, e.g. in the use of the hotspot-derived poles for tectonic reconstructions or for calibrating the APWP for dating purposes.

CONCLUSIONS

This study yields a ~36.5 Ma pole at 65.5°S, 112.5°E, consistent in latitude and longitude with four other poles that define the Palaeogene part of the Australian apparent polar wander trajectory. The four poles and a fifth reported elsewhere lie west of an APWP determined by curve fitting of scattered basaltic poles (Musgrave 1989). The scatter is likely to be due to the presence of uncancelled palaeosecular variations in the remanence directions. The new pole adds support for a systematic westward displacement of the

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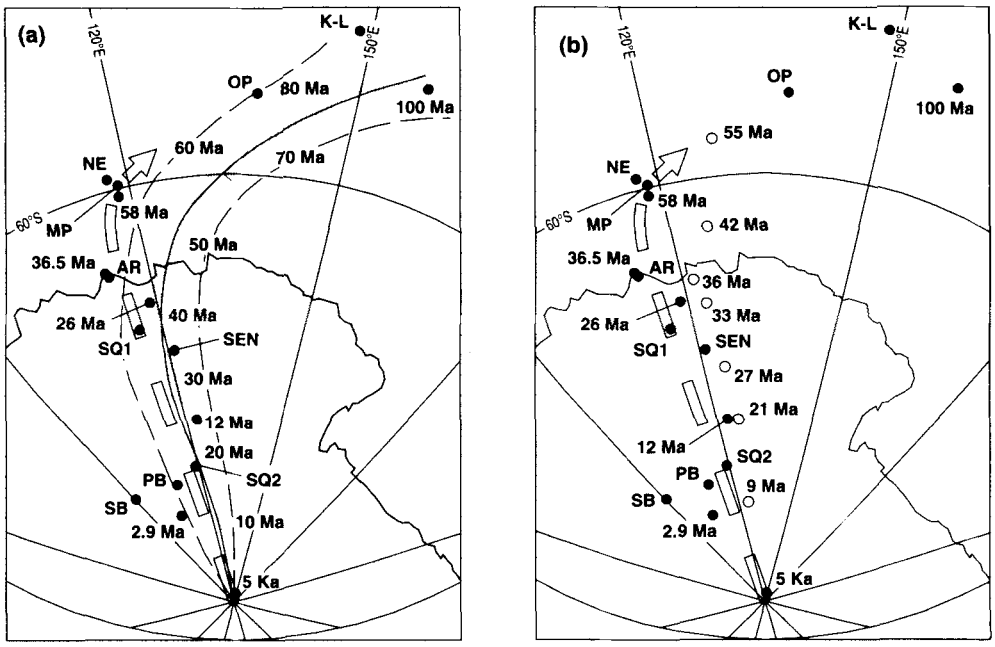


Figure 7. Comparison of the APWP of Fig. 5 (solid circles and open arrow) with (a) basalt-derived APWP of Musgrave (1989), and (b) poles derived from the hotspot model of Duncan (1981) (open circles). In Fig. 6(a) the shaded bars are confidence intervals of the age-calibration points (ages shown alongside) for the basalt-derived APWP, and the broken lines indicate the 95 per cent confidence envelope for its trajectory. Note a systematic westward deviation of the Palaeogene part of the present path from both the basalt-derived path and the poles obtained from the hotspot model.

Palaeogene palaeomagnetic poles from hotspot-derived poles. This difference may be due to true polar wander and/or non-dipole components in the geomagnetic field, two sources of second-order error when the palaeomagnetism and hot-spot volcanism are used as absolute palaeogeographic references.

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